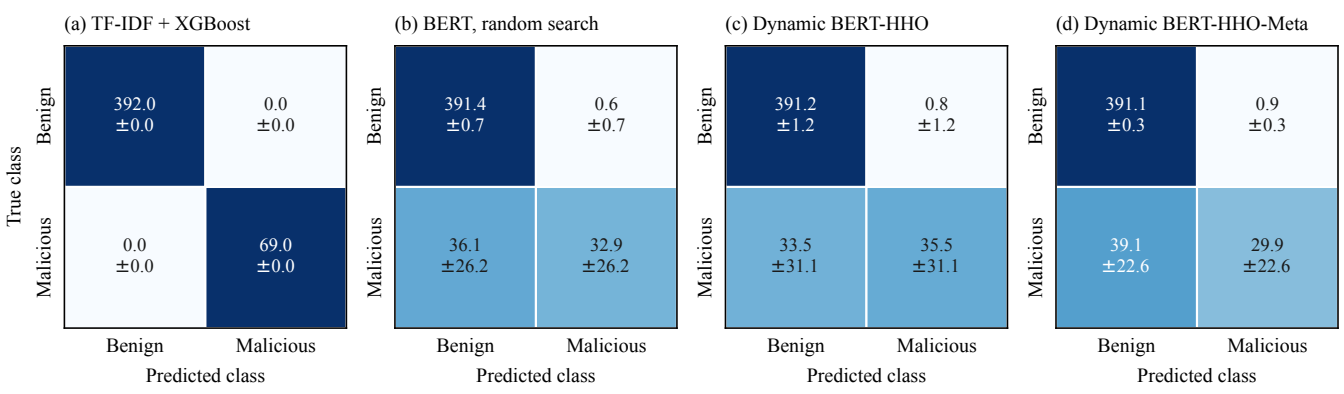
(a) TF-IDF + XGBoost
(b) BERT, random search
(c) Dynamic BERT-HHO
(d) Dynamic BERT-HHO-Meta
True class
Benign
Malicious
392.0 ±0.0
0.0 ±0.0
0.0 ±0.0
69.0 ±0.0
391.4 ±0.7
0.6 ±0.7
36.1 ±26.2
32.9 ±26.2
391.2 ±1.2
0.8 ±1.2
33.5 ±31.1
35.5 ±31.1
391.1 ±0.3
0.9 ±0.3
39.1 ±22.6
29.9 ±22.6
Predicted class